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GLOBAL IDEA

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The Oil Manual

'Let the Oil Flow'

MORGAN STANLEY & CO. INTERNATIONAL PLC+

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We adjust our forecast following the US/Iran MOU. We still see a tight summer ahead, and suspect Dated Brent can rebound from current levels into 3Q. However, with prospects of SoH recovery improved, our previous forecasts now appear too high. From 4Q onwards, we see Brent anchored around $80/b.

Key Takeaways

- Our previous modelling assumptions reflected a political agreement by late July; with the MOU announced over the weekend, this has come ~2 weeks earlier
- From here, it likely takes several weeks for tanker flow to be restored; we see 50% of production back by Sept, and 80% by Dec, slightly faster than before
- Despite the disruption, a broad range of indicators has signaled weakness in physical oil markets in recent weeks; notably, unsold cargos are above-average
- The 'twin solvers' of high US exports and low China imports are the key drivers; in the short term (i.e. next weeks) they do not seem to come to an end just yet
- In the end, we still see a tight summer that can support a modest rebound, but that period now appears too short to support our previous Brent forecast

Exhibit 1: For any given month, information on tanker movements firms up over time. Based on known voyages, it appears that net seaborne oil imports into China will decline even further in June vs May, and even July is tracking below May as well.

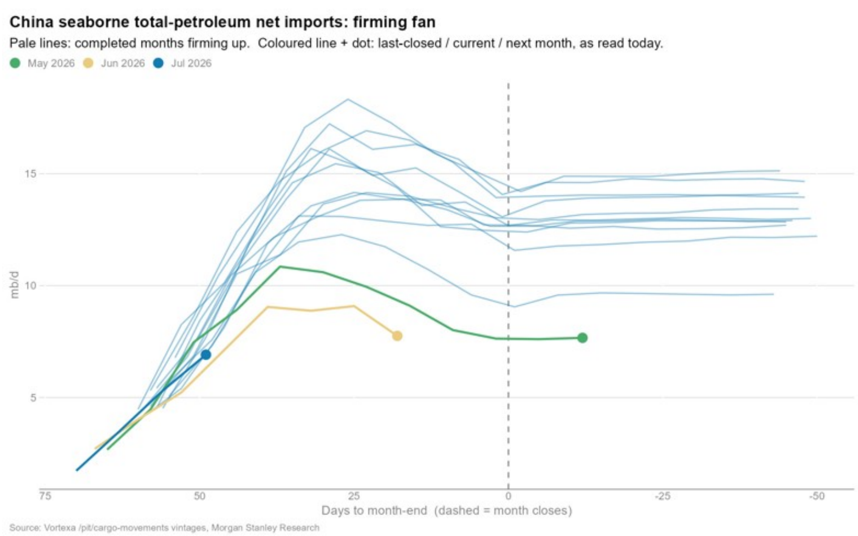


Exhibit 2 : We revise our near-term Brent price lower; our long-term forecast remains anchored around $80/b

Dated Brent price forecasts

($/bbl)	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27
New	90.0	80.0	80.0	80.0	80.0	80.0
Previous	100.0	95.0	85.0	80.0	80.0	80.0
Change	*−10.0*	*−15.0*	*−5.0*	*0.0*	*0.0*	*0.0*

Source: Morgan Stanley Research

Previous research:

- The Oil Manual: How Fast Can Middle East Production Return? (1 June 2026)
- The Oil Manual: SPR Tracker: How Much Has Been Released? What's Next? (28 May 2026)
- The Oil Manual: Why is the Price of Oil Not Higher? (11 May 2026)
- The Oil Manual: US Gasoline: Tight Balance; Fair Value For Now (4 May 2026)
- The Oil Manual: Schrodinger's Strait (27 Apr 2026)
- The Oil Manual: Hormuz Closure – Six Weeks In: Where Do We Stand? (13 Apr 2026)
- The Oil Manual: 'Paper' vs 'Physical' Oil - A Short Primer (7 Apr 2026)

'Let the Oil Flow'

Re-opening of the Strait agreed

106 days after the start of the conflict, President Trump announced yesterday that the United States and Iran have reached an intermediate agreement. Although the precise content of this MoU has not been released, both sides have indicated that it calls for 1) an end to hostilities on all fronts, 2) the opening of the Strait of Hormuz from both sides, and 3) a negotiation over Iran's nuclear programme and highly enriched uranium over a 60-day period.

In the end, this announcement came even slightly earlier than our modelling assumptions had implied. In our note How Fast Can Middle East Production Return? (1 Jun 2026), we had updated our estimate for the timing of a political break-through to 'late June'; it turned out to be 'mid-June' instead.

Much is still to be negotiated and key risks remain, but for now, this is a key step towards a de-escalation of the conflict and higher oil exports via the Strait of Hormuz.

Trajectory for production recovery brought forward by 1-2 weeks

Full restoration of tanker flow through the Strait will first require a period for the clearing of sea mines. There are alternative routes that avoid the traditional shipping lanes, but the capacity of these routes is likely only a fraction of pre-conflict transits.

Then, some time is likely needed to rebuild commercial confidence amongst ship owners and insurance companies, and allow tankers - many of which have relocated elsewhere - to come back to the region.

Also, for production to be restored, export tanks need to be cleared first, which means that the pace at which empty tankers *enter* the Gulf is arguably even more important than laden tankers leaving.

For these reasons, we estimated that the ramp-up in production would start from late-July, which we now bring forward to mid-July . After that, we assume that 50% of lost production will be back by September, 80% by December, with the rest to follow in early 2027.

Exhibit 3: Outbound tanker transit are still sharply below pre-conflict levels but notably improved since early May

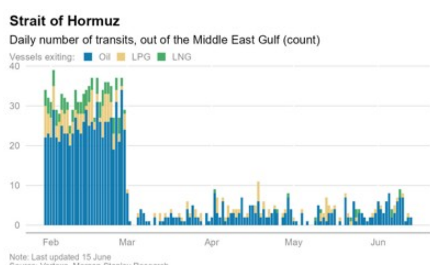
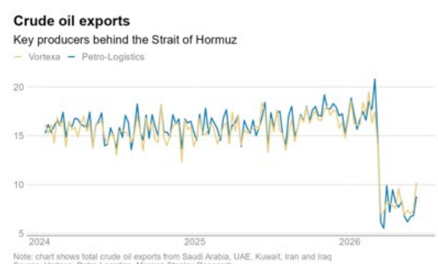


Exhibit 4: With that, also oil exports from the region have seen a small but notable bounce from the low



Despite ongoing supply loss, physical market has softened

By now, the cumulative supply loss from the Middle East since 1 March has reached ~1.4 billion barrels across both crude oil and refined products, relative to the same period in 2025. Nevertheless, the Brent market has softened visibly in recent weeks.

This has not only been a story of Brent futures anticipating the Strait to re-open. Dated Brent has come under pressure too, both the DFL and CFD curves have weakened in both level and structure, physical diffs have softened, the cash Dubai premium has been on a steady decline, and refined product cracks have mostly been on a downward slope too. The stand-out here is the naphtha crack spread in Asia relative to Dubai crude; this has fallen $13/b in the last four weeks to $(11)/b, close to its 1-year low - quite a counter-intuitive weakening for a production for which 33% of seaborne supply has been cut off by the Strait of Hormuz.

This weakening can also be seen in the overview of unsold cargoes. The table below summarises indications of unsold cargoes over the last two weeks. To be clear, the oil industry always has some cargoes that struggle to sell - it's part of the friction of buying and selling. However, we would characterise the current level as higher-than-normal and broadening. And that would be compared against 'normal' times. Considering that 11 mb/d of crude oil production is currently shut-in across the Middle East, this is quite unusual and highlights the physical weakness in the market.

Exhibit 5: We'd characterise the current number of unsold cargoes as above-normal and broadening in recent weeks

Unsold crude cargoes - latest estimate by region/grade

As of 15 June 2026. Trader anecdotes, not a continuous series.

Region / grade	Unsold estimate	As of	Source
Nigeria	July programme "largely unsold"; Bonny Light & Forcados "numerous unsold cargoes"; ~15mn bl June unplaced earlier in cycle	Jun 12 / Jun 5	Platts / Argus
Angola	~a dozen stems (10-12 cargoes) July; ~10mn bl July explicitly unsold	Jun 12 / Jun 10	Platts / Argus
Congo (Djeno)	Some July cargoes unsold at ~$4/bl under Dated - "not attractive enough for Chinese refiners"	Jun 10	Argus
CPC Blend	June 2-3 cargoes; rebounded off Jan-2023 low (Jun 11) but July "plenty unsold first decade", Asia arb unviable; 11-Suezmax injection lingering	Jun 9-11	Platts
Libya / Es Sider	7-10 June cargoes (incl. 4-5 Es Sider); unplaced Libyan barrels persist into July trade	Jun 5 / Jun 12	Platts / Argus
Azeri BTC / Kirkuk	Kirkuk unsold in June; Socar/Eni CPC offers went unplaced; first Basrah offers reaching Europe	Jun 3-5	Platts / Argus
North Sea	Forties ~1/3 of 9 June cargoes (May 26) - since cleared; Atlantic overhang still capping diffs & US exports	May 26 - Jun 2	Platts
Mideast Gulf	Spot thin but placing July cargoes - cleared	May 26	Platts
Russian ESPO	>50% of July programme unsold; June cargoes still unsold	Jun 12 / Jun 8	Argus / Platts
Brazil	Unsold July-delivery to China (late May); Aug cycle ~18mn bl taken - lower than prior	May 25-29	Argus
Guyana	July cycle "quiet", buying limited into third week; June stragglers still offered	Jun 11	Argus
Canada / TMX	August injection cargoes "still to be sold", count unclear	Jun 12	Argus
Asia-Pac condensate	July NWS unsold, pressuring August values & cracks	Jun 3-5	Argus

Source: Platts, Argus, Morgan Stanley Research

High US exports, low Chinese imports...but little change expected in June

The reasons for this are the same as those discussed in prior research: unusually large exports from the United States, unusually low imports into China, and weak apparent demand.

The charts below show the first two. In rounded numbers, net exports of crude oil and refined products have increased from ~5 mb/d last year to nearly ~9 mb/d recently. At the same time, China's net imports of crude and products have declined from ~13 mb/d last year to ~7 mb/d recently. With ~4 mb/d more from the US, and ~6 mb/d less going into China, these two countries have shielded the rest of the world - including the Brent market - of 10 mb/d of tightness.

Exhibit 6: US seaborne net exports are nearly 4 mb/d higher than last year...

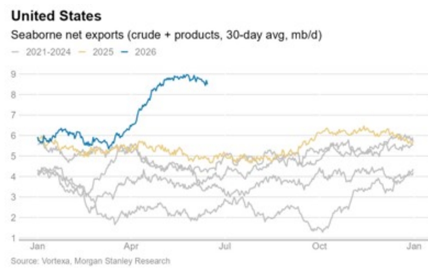
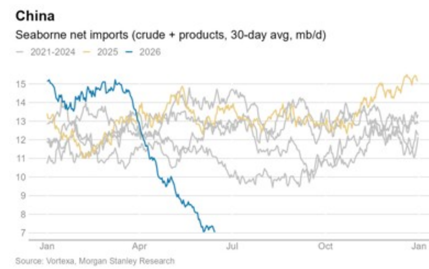


Exhibit 7: ...whilst China's seaborne net imports are down ~ 6mb/d

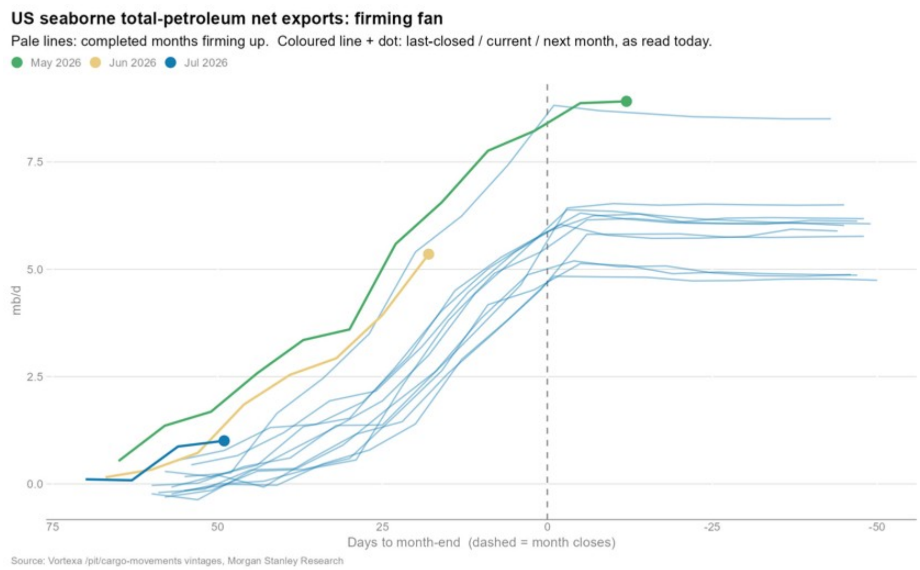


It is unlikely, however, that both countries can sustain this. The key question is, of course, when will this come to an end? We previously mentioned that if US exports were to fall and/or China's imports were to increase before the Strait of Hormuz was open, the Brent market could have a period of strength again. That may still happen, but it has become less likely.

We can estimate US exports and China imports based on known future tanker movements. Naturally, fewer tanker movements are known further in advance, but they 'firm up' as time goes by. The two exhibits below show the 'firming fans' for both time series - they show how known tanker movements related to US exports and China imports have developed over time.

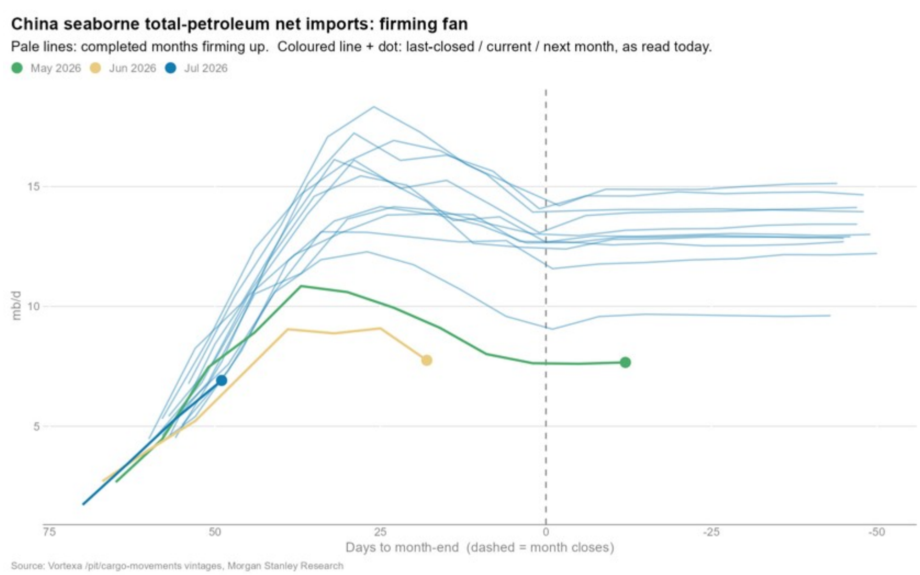
US net exports for June 2026 are currently tracking the path of April, so is still likely to end up in the 8.5-9 mb/d range, broadly where the May level ended as well. It is still early to judge July exports on this basis, which are still tracking below both the May and June trajectories so far, but are still above any of the previous 12 months.

Exhibit 8: Based on the pace at which known tanker movements firm up, seaborne net exports from the US are tracking similar to April, which is also where May levels ended up



Known tanker movements for June imply even lower net imports into China than for May, and data for July is also still tracking below the May path so far:

Exhibit 9: Along similar lines, known tanker movements in/out of China suggest that June net imports will likely be even lower than May, and July is also tracking below the May level as well

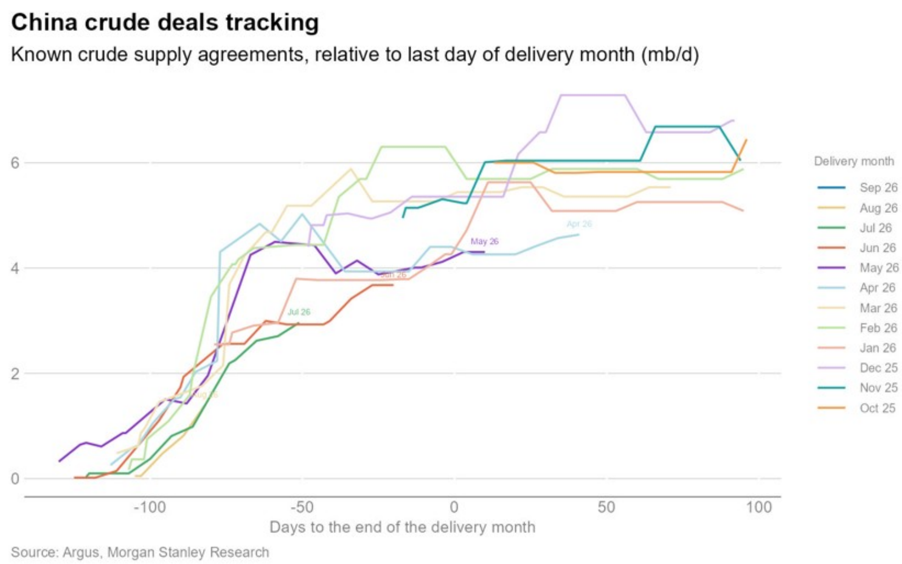


In addition to this, we can track China's buying activity by aggregating individual spot crude deals entered into by Chinese refiners, trading houses and other companies. Spot deals usually only cover ~6 mb/d out of ~11 mb/d of crude oil imports; the remainder comes from deliveries under longer-term contracts.

Spot deals supplied an unusually low ~4.5 mb/d in April, and an even lower ~4.1 mb/d in

May. So far, known-deals for June are tracking even lower than the paths for April and May, and data for July and August shows little to no improvement either:

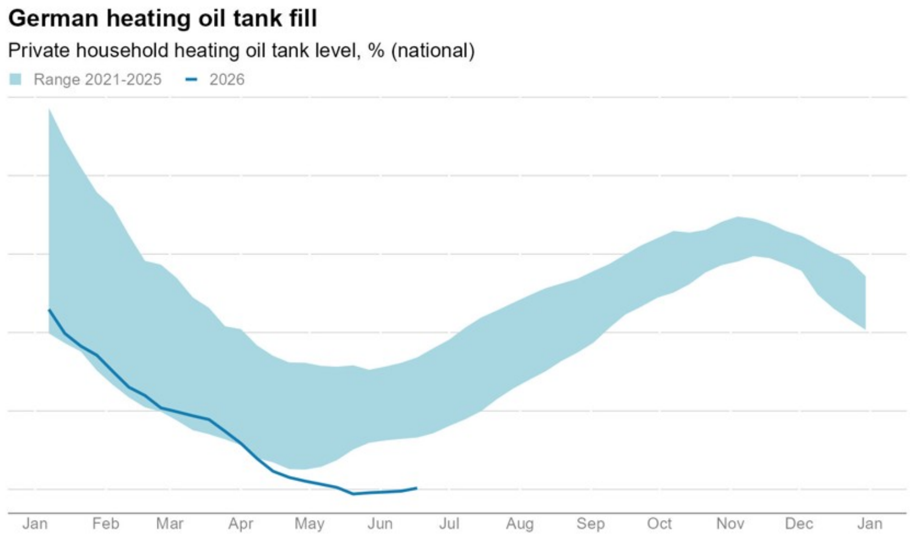
Exhibit 10: Known spot crude deals entered into by Chinese players do not signal a pickup in buying interest yet



Finally, on demand. We estimate that global refinery crude runs were down ~5.8 mb/d year-on-year in May, and with this, global crude demand was down by the same. This too has had a significant impact on crude oil prices.

However, our strong suspicion is that this sharply overstates the underlying demand decline. Instead, this decline in crude demand - and the associated decline in refined products production - is largely driven by delayed purchasing, not necessarily lower consumption. We have limited data to describe this effect fully on a global basis, but Exhibit 11 provides an example:

Exhibit 11: Expecting the Strait of Hormuz to open soon, German households have not been filling their heating oil tanks - consumption has continued but buying is delayed



Source: Argus MDX Monitor, Morgan Stanley Research

Exhibit 11 shows data on the average fill rate of heating oil tanks owned by German households. In total, approximately 4 million German households use heating oil (which is practically identical to diesel) and have their own storage facility for this. In total, their tank capacity is close to 100 million barrels - i.e. not small.

The fill rate of these tanks typically follows a seasonal pattern - they draw over the winter, and build in the summer. However, this has not been the case recently. By now, German households would typically have started to refill storages ahead of next winter but they have not done so yet in 2026. The most likely reason: they too have been waiting from the Strait of Hormuz to re-open, expecting to buy at lower prices later.

In short: they continue to consume, but they are not buying. In the end, 'buying' is what matters for oil prices. Data on tertiary inventories (i.e. inventories held by end-consumers directly) is notoriously limited; this is the only real example we have. However, given that the re-opening of the Strait has been imminent at any point over the last 3 months, our strong suspicion is that the buying-vs-consumption pattern of German heating oil consumers has been symptomatic for many other part of the oil complex.

Putting it all together - still a tight summer, but prices likely remain lower than estimates before

In the end our balances still show a tight summer. For 3Q, we estimate a deficit of 3.4 mb/d, with observable stocks likely drawing 2.1 mb/d, of which 1.1 mb/d in the OECD (which weighs heavier in price formation). However, with yesterday's MOU announcement, this undersupply is likely to remain limited to 3Q - that's it. With our current supply recovery assumptions, the oil market is likely to be back in balance in 4Q again.

We've previously estimated that in a balanced market with inventories at average levels, Brent is likely anchored around ~$80/b. Therefore, that is what we pencil in for 4Q (down from $95/b before).

Exhibit 12: We lower our Brent forecast for the next few quarters

Dated Brent price forecasts

($/bbl)	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27
New	90.0	80.0	80.0	80.0	80.0	80.0
Previous	100.0	95.0	85.0	80.0	80.0	80.0
Change	*−10.0*	*−15.0*	*−5.0*	*0.0*	*0.0*	*0.0*

Source: Morgan Stanley Research

Given the market is still likely in a considerable deficit in 3Q, we suspect Dated Brent prices will average above that level over the summer, especially as demand is likely to firm up from here. However, given that the 'twin-solvers' of high US exports and low Chinese imports seem to continue for now, the spread over $80/b can only be so much. With that in mind, we peg our average 3Q Dated Brent forecast at $90/b, down from $100/b before.

In our experience, 'next year's balance' has a habit of looking soft in the middle of the year before - and so it is again this year. In the end, there is often an unexpected supply outage somewhere that changes this outcome, but modelling the 'unknown unknowns' is not straightforward. Based on our current assumptions, the oil market returns to a surplus again in 2027, with observable stocks building 2.4 mb/d.

If $80/b is the reference price for a balanced market, in principle this suggests prices lower than that. However, there are offsetting effects. We now know that the Strait of Hormuz can be closed, and how much oil will flow through the Strait next year exactly is still unknown. Also, more inventory filling may be required. Based on heating oil inventories held by German households, our suspicion is that tertiary inventories have drawn significantly. On top of that, several countries are likely to expand/start building SPRs - and building SPRs acts like 'demand'. With that in mind, we stick to a $80/b Brent forecast for 2027, which implies a small downgrade to our 1Q 27 forecast (which was $85/b).

Supply/demand balance

Exhibit 13:

Total oil liquids	2025	1Q26	2Q26	3Q26	4Q26	2026	1Q27	2Q27	3Q27	4Q27	2027	*Change* '25-26	'26-27
Demand	**104.9**	**104.6**	**101.0**	**103.8**	**105.5**	**103.7**	**104.8**	**105.1**	**106.3**	**107.3**	**105.9**	**-1.2**	**2.1**
OECD	45.9	45.7	44.7	45.9	45.8	45.5	45.2	45.3	46.0	46.0	45.6	-0.3	0.1
Non-OECD	59.0	59.0	56.3	57.9	59.7	58.2	59.6	59.8	60.3	61.2	60.2	-0.9	2.0
Non-OPEC supply	**72.2**	**71.7**	**72.0**	**73.4**	**73.7**	**72.7**	**74.0**	**74.2**	**74.7**	**75.2**	**74.5**	**0.5**	**1.8**
United States C+C	13.6	13.6	14.0	13.9	14.0	13.9	14.1	14.1	14.1	14.1	14.1	0.3	0.2
FSU C+C	10.3	10.0	10.0	10.3	10.3	10.2	10.3	10.2	10.2	10.3	10.2	-0.1	0.1
Other	48.3	48.1	48.0	49.2	49.3	48.7	49.6	49.9	50.4	50.7	50.2	0.3	1.5
OPEC NGLs and condensate	5.5	5.2	3.7	4.7	5.1	4.7	5.3	5.4	5.4	5.4	5.4	-0.9	0.7
Call on OPEC crude	**27.2**	**27.7**	**25.4**	**25.7**	**26.7**	**26.4**	**25.6**	**25.6**	**26.3**	**26.7**	**26.0**	**-0.9**	**-0.3**
OPEC crude	29.6	27.3	19.6	22.3	27.4	24.2	29.2	30.0	30.4	30.5	30.0	-5.5	5.8
Implied balance	**2.4**	**-0.4**	**-5.8**	**-3.4**	**0.7**	**-2.2**	**3.7**	**4.4**	**4.1**	**3.8**	**4.0**	**-4.6**	**6.2**
Misc to bal	-1.2	-1.0	3.8	1.3	-0.3	0.9	-1.5	-1.8	-1.7	-1.5	-1.6	2.1	-2.6
Observed stock change	**1.2**	**-1.4**	**-2.0**	**-2.1**	**0.4**	**-1.3**	**2.2**	**2.6**	**2.5**	**2.2**	**2.4**	**-2.5**	**3.6**
of which:													
SPR (IEA members)	0.0	-0.1	-2.0	-1.0	-	-0.8	0.5	0.5	0.5	0.5	0.5	-0.8	1.3
China	0.3	0.6	-0.1	-	-	0.1	0.3	0.3	0.3	0.3	0.3	-0.2	0.2
Other non-OECD	0.1	0.2	-	-	-	0.1	0.3	0.3	0.3	0.3	0.3	-0.1	0.2
Commercial OECD	**0.1**	**-0.1**	**-0.0**	**-1.1**	**0.4**	**-0.2**	**1.1**	**1.5**	**1.4**	**1.1**	**1.3**	-0.3	1.5
Oil-on-water	0.6	-2.0	0.1	-	-	-0.5	-	-	-	-	-	-1.1	0.5

Crude + condensate only	2025	1Q26	2Q26	3Q26	4Q26	2026	1Q27	2Q27	3Q27	4Q27	2027	*Change* '25-26	'26-27
Runs	**84.2**	**83.6**	**78.1**	**82.5**	**84.6**	**82.2**	**84.9**	**84.7**	**85.6**	**85.3**	**85.2**	**-2.0**	**3.0**
West of Suez	43.3	42.9	42.9	44.0	43.2	43.2	42.8	43.1	43.9	43.2	43.2	-0.1	-0.0
East of Suez	40.9	40.6	35.2	38.5	41.3	38.9	42.1	41.7	41.7	42.2	41.9	-2.0	3.0
Direct use	**0.9**	**0.7**	**0.9**	**1.0**	**0.6**	**0.8**	**0.6**	**0.8**	**1.0**	**0.6**	**0.8**	**-0.1**	**-0.0**
Non-OPEC C+C supply	**54.6**	**54.5**	**54.5**	**55.2**	**55.7**	**55.0**	**56.1**	**56.0**	**56.2**	**56.7**	**56.3**	**0.3**	**1.3**
United States	13.6	13.6	14.0	13.9	14.0	13.9	14.1	14.1	14.1	14.1	14.1	0.3	0.2
Other	41.1	40.9	40.5	41.3	41.7	41.1	42.1	41.9	42.1	42.6	42.2	0.0	1.1
OPEC	**31.6**	**29.1**	**20.9**	**23.9**	**29.2**	**25.8**	**31.0**	**31.8**	**32.2**	**32.2**	**31.8**	**-5.9**	**6.0**
Crude	29.6	27.3	19.6	22.3	27.4	24.2	29.2	30.0	30.4	30.5	30.0	-5.5	5.8
Condensate	2.0	1.8	1.3	1.6	1.7	1.6	1.8	1.8	1.8	1.8	1.8	-0.4	0.2
Implied stock change	**1.2**	**-0.7**	**-3.6**	**-4.3**	**-0.3**	**-2.2**	**1.5**	**2.2**	**1.8**	**3.0**	**2.1**	**-3.4**	**4.4**
Dated Brent forecasts (period averages)													
Base	**69.1**	**75.7**	**110**	**90**	**80**	**89**	**80**	**80**	**80**	**80**	**80**		
Bull	69.1	75.7	110	120	120	106	120	120	120	120	120		
Bear	69.1	75.7	110	75	65	81	65	65	65	65	65		
WTI													
Base	**64.7**	**71.4**	**102**	**84**	**75**	**84**	**75**	**75**	**75**	**75**	**75**		
Bull	64.7	71.4	145	145	145	145	145	145	145	145	145		
Bear	64.7	71.4	106	71	61	77	61	61	61	61	61		

Source: Argus, IEA, EIA/DOE, Vortexa, S&P Global Energy, Petro-Logistics, Rystad Energy, Morgan Stanley Research

Key market indicators

Exhibit 14:

Global oil prices

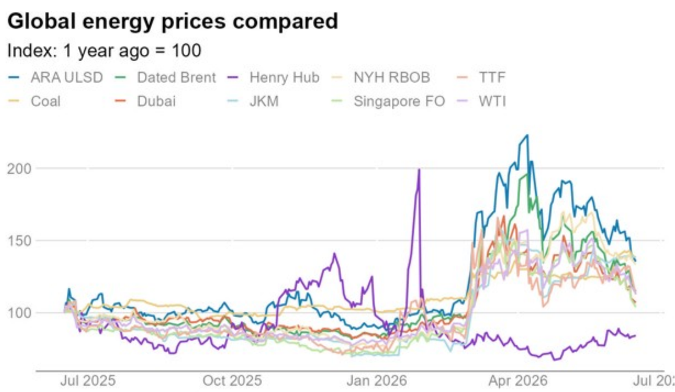
Latest value as per 15 Jun 2026

Instrument	Latest	Change 1 wk	Change 4 wk	Unit	Last 1m	1Y low	1Y range	1Y high	Rationale for including
Global benchmarks									
ICE Brent M1 future (Aug)	83.2	−11.0	−28.9	$/bbl		58.9		118.3	Global crude benchmark; anchor for Atlantic pricing
NYMEX WTI M1 future (Jul)	80.5	−10.8	−28.2	$/bbl		55.3		113.0	US marginal barrel; key global risk benchmark
Dubai M+2 swap	78.9	−14.7	−29.0	$/bbl		58.2		169.8	Asia sour benchmark; core signal for Middle East flows
Term structure									
ICE Brent future M1-M2 (Aug/Sep)	0.96	−1.0	−3.7	$/bbl		−0.39		14.38	Prompt tightness/storage signal in Brent complex
ICE Brent future M2-M3 (Sep/Oct)	1.0	−1.0	−3.7	$/bbl		0.0		8.0	Cleaner structure signal when M1 rolls/noisy
NYMEX futures M1-M2 (Jul/Aug)	1.4	−0.7	−2.9	$/bbl		0.0		13.9	US prompt balance signal; inventories & pipeline pull
NYMEX futures M2-M3 (Aug/Sep)	1.2	−1.1	−3.7	$/bbl		0.0		8.6	Back-end prompt signal; reduces expiry distortion
Dubai swap M1-M2	0.25	−1.8	−3.8	$/bbl		−0.11		14.25	Asia prompt balance; sour crude tightness indicator
Dubai swap M2-M3	0.33	−1.0	−2.8	$/bbl		−0.10		6.11	Structure beyond front; filters front-month noise
North Sea									
Dated Brent	88.2	−9.6	−24.0	$/bbl		60.2		144.4	Physical spot benchmark; sets Atlantic Basin crude value
Brent DFL (M1)	1.2	−1.9	−4.5	$/bbl		−0.7		13.8	Bridge between Dated and futures; spot vs paper alignment
Brent CFD (Week 1)	2.2	−3.5	−4.7	$/bbl		−1.2		43.9	Very prompt NS tightness; near-term cargo availability
Forties vs Dated Brent strip	1.0	0.1	−0.3	$/bbl		−1.1		21.2	Marginal sour North Sea grade; barometer for basin length
Ekofisk vs Dated Brent strip	3.6	0.0	0.7	$/bbl		0.6		22.6	Sweet premium proxy; refinery demand for light/sweet yield
J. Sverdrup vs Dated Brent strip	3.7	1.3	1.2	$/bbl		−2.9		16.7	Medium sour NS value; competes with ME grades
United States									
WTI Cushing M1 (cash)	84.9	−5.7	−20.5	$/bbl		55.3		113.0	Physical WTI anchor at Cushing; pipeline hub clearing price
WTI Midland vs WTI Cushing	0.25	−0.3	−0.8	$/bbl		−0.35		4.50	Permian takeaway stress signal; pipes vs production growth
WTI MEH vs WTI Cushing	0.50	−0.3	−1.0	$/bbl		−0.35		7.75	Export outlet premium/discount; USGC pull vs inland
WCS Hardisty vs WTI Cushing CMA	−11.5	0.5	4.3	$/bbl		−16.9		−9.4	Heavy-light & Canada logistics; refinery margin barometer
WTI-Brent spread	−4.1	0.9	3.6	$/bbl		−17.0		0.5	Key east-west arb signal; drives US export economics
Middle East									
Brent-Dubai EFS (M1)	6.5	−2.9	−6.3	$/bbl		−0.3		18.5	Atlantic vs Asia arb; key driver of Brent vs Dubai relative value
Cash Dubai premium	2.3	−4.5	−7.9	$/bbl		−0.6		65.7	Spot tightness vs swaps; immediate OSP pressure signal
Oman vs Dubai (M1)	2.3	−4.5	−7.9	$/bbl		−0.3		65.7	ME benchmark for China; complements Dubai for sour/light
Murban Quality Premium	0.00	0.0	0.0	$/bbl		−34.54		1.11	Measures Murban's quality value vs Dubai benchmark
West Africa									
Kissanje vs Dated Brent strip	−3.0	0.0	−0.1	$/bbl		−4.9		10.1	China-bound WAF barrel; marginal Atlantic supply signal
Jubilee vs Dated Brent strip	−1.5	0.0	0.0	$/bbl		−2.7		10.0	Atlantic sweet clearing level; tracks China/Europe demand
Hungo vs Dated Brent strip	−4.5	0.0	0.0	$/bbl		−6.2		11.3	Medium sweet WAF value; sensitive to China buying swings
Djeno vs Dated Brent strip	−6.2	0.2	0.6	$/bbl		−8.2		10.6	Sour WAF grade; key China-bound arb and freight sensitivity
Long-haul arbitrage grades									
WTI FOB USGC vs Dated Brent strip	−1.1	0.6	1.2	$/bbl		−11.0		2.0	US export competitiveness vs Brent; arb opens/closes
Guyana Liza vs Dated Brent strip	−1.4	0.0	−0.3	$/bbl		−3.9		8.3	Growth light sweet barrel; clears into China/Europe
Tupi (Brazil) vs Dated Brent strip	−2.5	−0.5	−1.5	$/bbl		−10.6		9.4	Core Brazil export arb; long-haul pull indicator
CPC Blend vs Dated Brent strip	−9.5	1.7	−3.4	$/bbl		−11.2		0.0	Black Sea arb barrel; freight/geopolitics sensitive
Freight									
Platts Global VLCC TCE Index	268.8	3.7	−24.4	k$/day		18.1		519.1	Single global crude freight gauge; ton-mile weighted signal
Persian Gulf → China (VLCC)	73.5	0.0	−5.7	$/mt		9.5		124.1	Main ME→Asia flow cost; drives Dubai/Murban economics
US Gulf Coast → China (VLCC)	61.1	1.9	−3.7	$/mt		24.3		108.5	Key cost for US exports; links WTI FOB to Asia clearing
West Africa → China (VLCC)	44.1	4.4	−4.2	$/mt		17.1		95.0	Cost for China-bound Atlantic grades; clears WAF differentials
Refined products									
Asian Naphtha crack vs Dubai	−11.1	1.4	−12.7	$/bbl		−13.8		27.4	Petchem demand gauge; Asia light-end support signal
USGC Gasoline crack vs WTI	33.6	4.0	−7.2	$/bbl		10.8		43.1	Gasoline margin driver; swing refining runs and crude demand
Spore Jet crack vs Dubai	35.0	−11.4	−16.5	$/bbl		15.8		114.2	Asia travel demand proxy; middle distillate strength signal
ICE Gasoil crack vs Brent	41.5	−5.9	−14.5	$/bbl		20.4		69.4	Global distillate benchmark margin; key crude demand driver
Spore 380 CST FO crack vs Dubai	0.29	−3.0	−6.2	$/bbl		−9.45		14.04	Residual value signal; heavy sour crude support/drag

Source: Argus, Bloomberg, Platts, General Index, Morgan Stanley Research

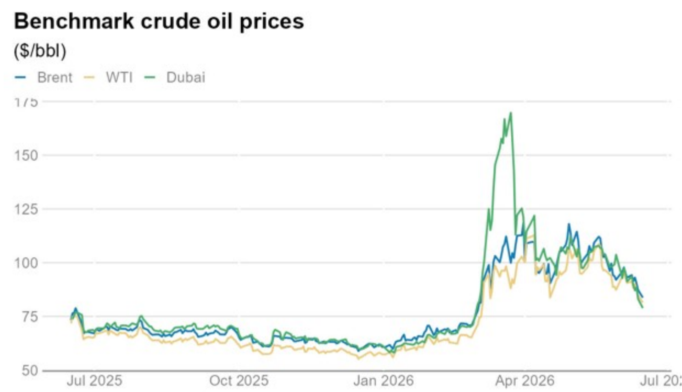
Prices

Exhibit 15:



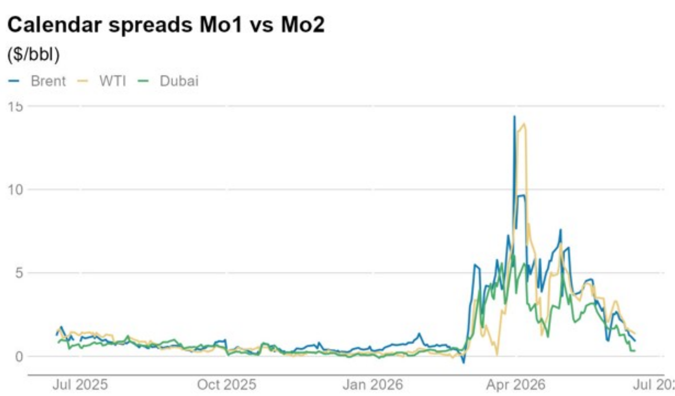
Source: Bloomberg, Platts, Morgan Stanley Research

Exhibit 16:



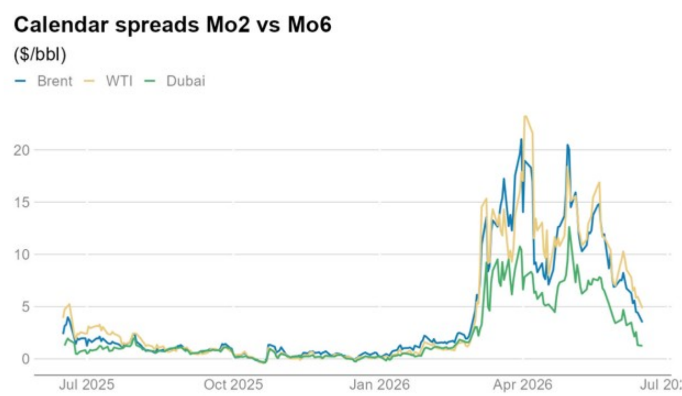
Source: Bloomberg, Platts, Morgan Stanley Research

Exhibit 17:



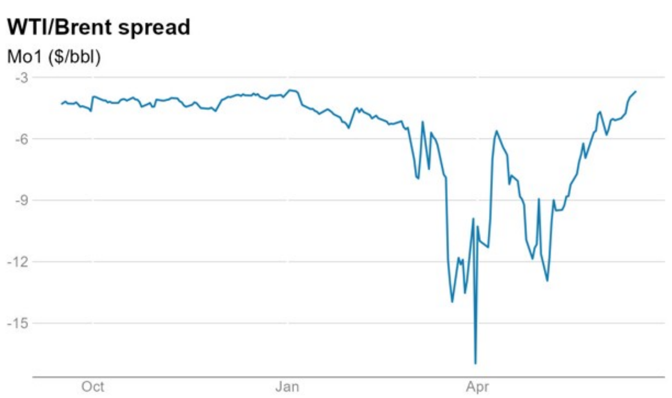
Source: Bloomberg, Platts, Morgan Stanley Research

Exhibit 18:



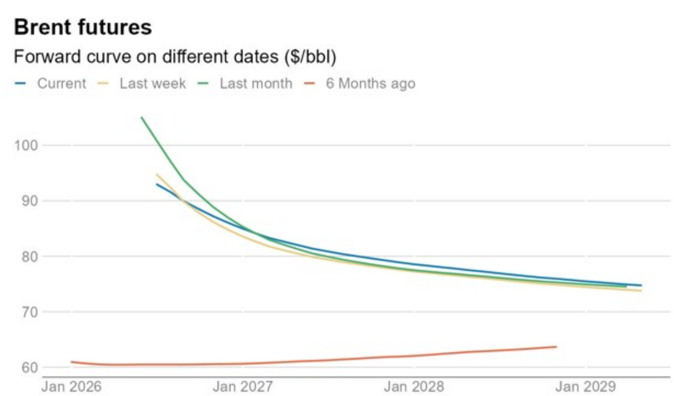
Source: Bloomberg, Platts, Morgan Stanley Research

Exhibit 19:



Source: Bloomberg, Morgan Stanley Research

Exhibit 20:

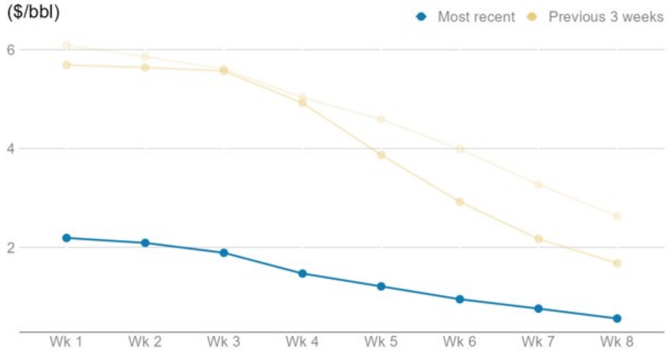


Source: Platts, Morgan Stanley Research

Exhibit 21:

Brent CFD curve

($/bbl)

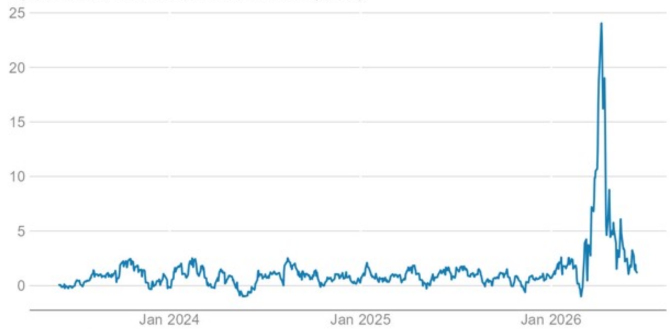


Source: S&P Global Platts, Morgan Stanley Research

Exhibit 22:

CFD curve structure

Spread between week 2 and week 6 ($/bbl)

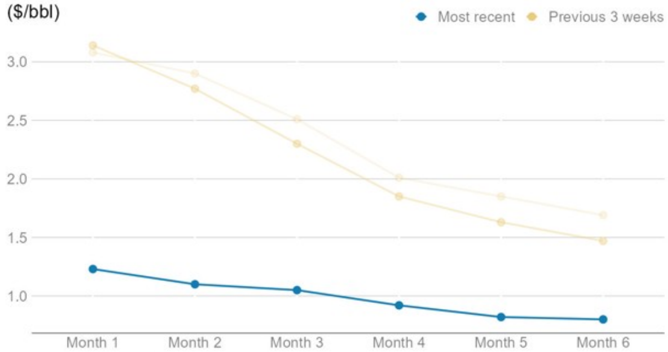


Source: S&P Global Platts, Morgan Stanley Research

Exhibit 23:

Brent DFL curve

($/bbl)



Source: S&P Global Platts, Morgan Stanley Research

Exhibit 24:

DFL curve structure

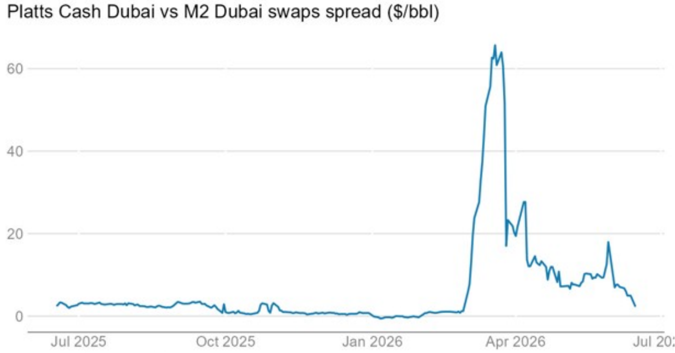
Spread between month 1 and month 6 ($/bbl)

Source: S&P Global Platts, Morgan Stanley Research

Exhibit 25:

Cash Dubai premium

Platts Cash Dubai vs M2 Dubai swaps spread ($/bbl)

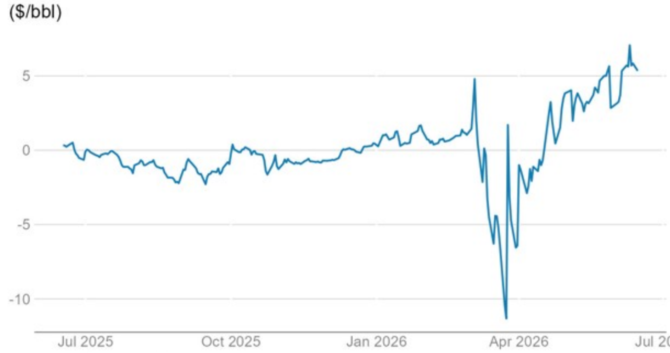


Source: S&P Global Platts, Morgan Stanley Research

Exhibit 26:

Brent/Dubai spread

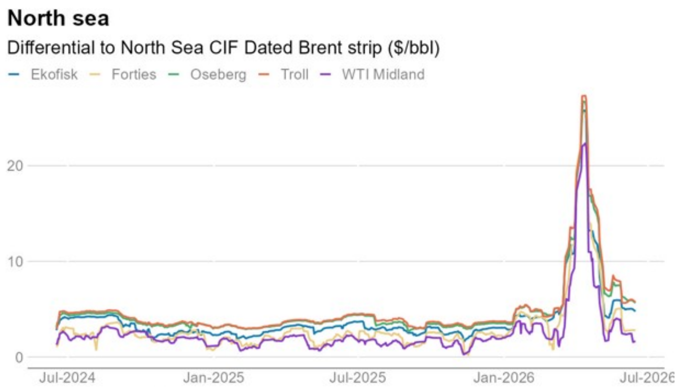
($/bbl)



Source: S&P Global Platts, Morgan Stanley Research

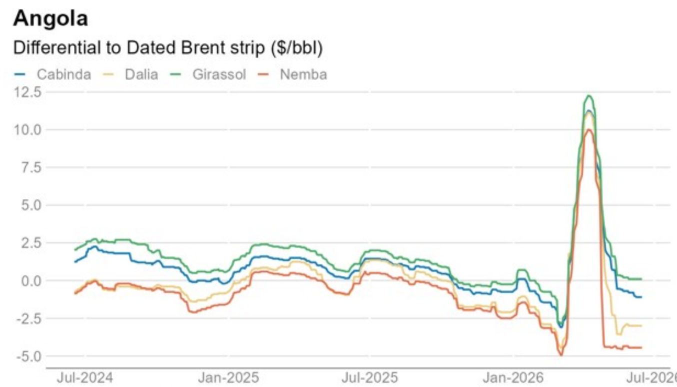
Differentials

Exhibit 27:



Source: S&P Global Platts, Morgan Stanley Research

Exhibit 28:

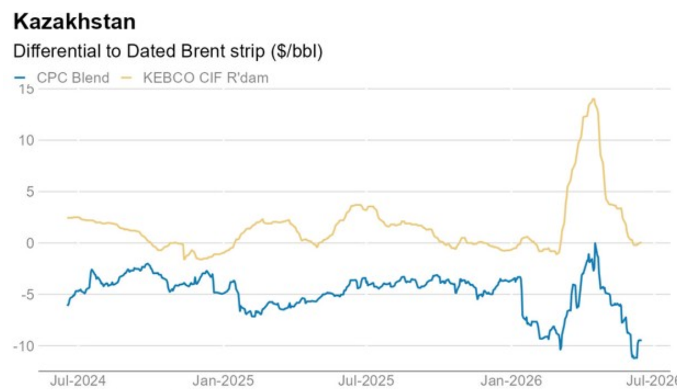


Source: S&P Global Platts, Morgan Stanley Research

Exhibit 29:

Source: S&P Global Platts, Morgan Stanley Research

Exhibit 30:

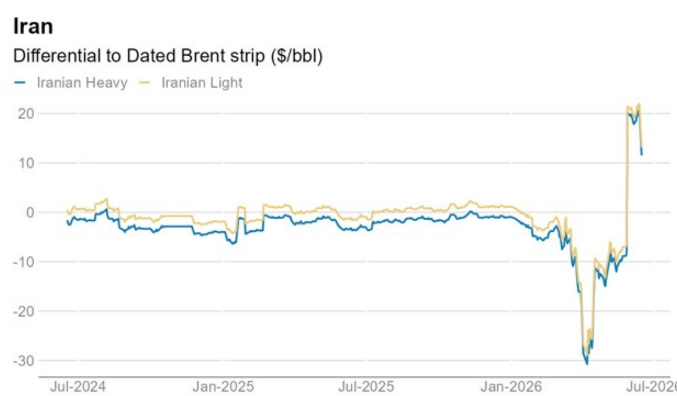


Source: S&P Global Platts, Morgan Stanley Research

Exhibit 31:

Source: S&P Global Platts, Morgan Stanley Research

Exhibit 32:



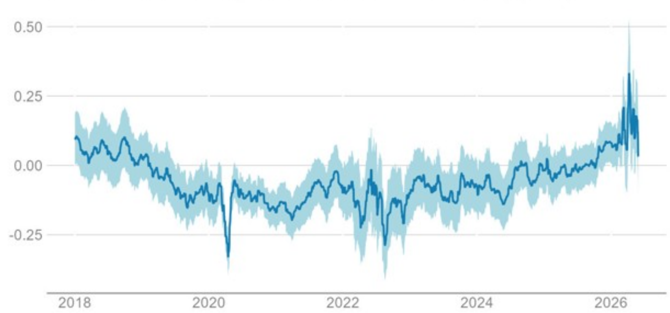
Source: S&P Global Platts, Morgan Stanley Research

Crude Quality

Exhibit 33:

Heavy vs Light

Premium/(discount) for every 1 degree increase in API Gravity ($/bbl)

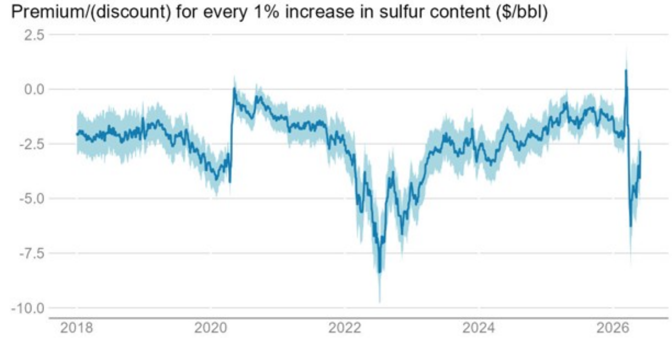


Note: analysis based on weekly regression across 95 seaborne crudes
Source: Platts, Morgan Stanley Research

Exhibit 34:

Sweet vs Sour

Premium/(discount) for every 1% increase in sulfur content ($/bbl)

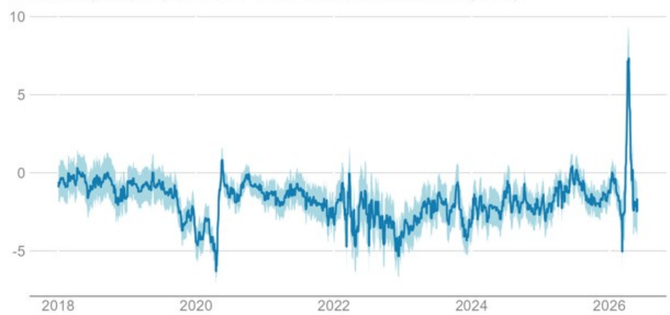


Note: analysis based on weekly regression across 95 seaborne crudes
Source: Platts, Morgan Stanley Research

Exhibit 35:

East vs West of Suez

Premium/(discount) for West- over East-of-Suez crudes ($/bbl)

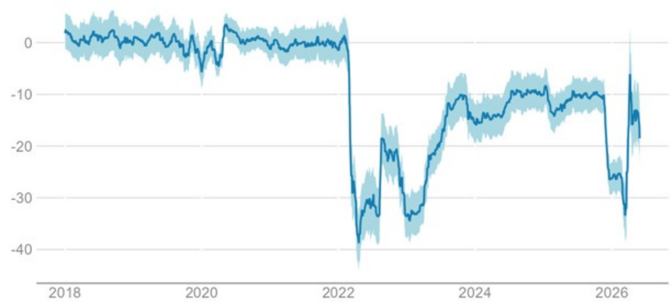


Note: analysis based on weekly regression across 95 seaborne crudes
Source: Platts, Morgan Stanley Research

Exhibit 36:

Russian vs non-Russian origin

Premium/(discount) for Russian crudes ($/bbl)



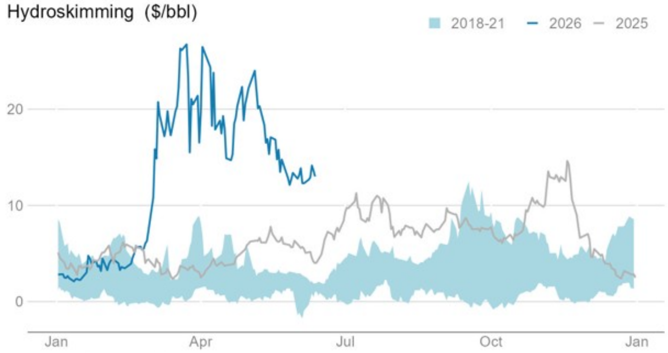
Note: analysis based on weekly regression across 95 seaborne crudes
Source: Platts, Morgan Stanley Research

The analyses above are based on a daily regression of the price of 95 seaborne crudes against 1) their API gravity, 2) their sulphur content, 3) a dummy variable indicating whether the crude's origin is East or West of Suez, and 4) another dummy variable indicating whether the grade is from Russian origin or not. The charts above show how the sensitivity of these 95 crude prices to each of these variables has changed over time.

Refining Margins

Exhibit 37:

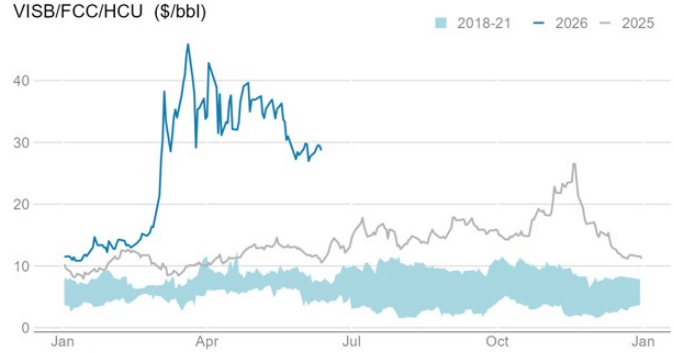
Cracking netback margin - Northwest Europe
Hydroskimming ($/bbl)



Source: S&P Global Platts, Morgan Stanley Research

Exhibit 38:

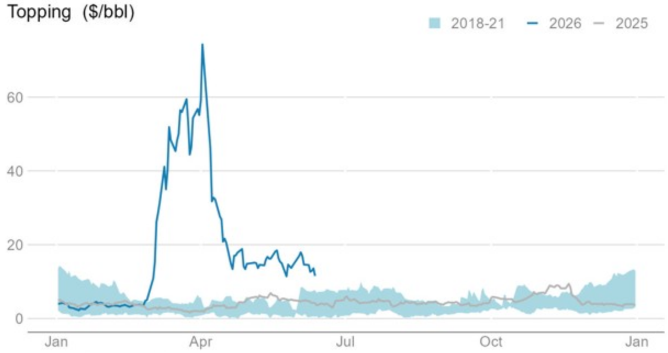
Cracking netback margin - Northwest Europe
VISB/FCC/HCU ($/bbl)



Source: S&P Global Platts, Morgan Stanley Research

Exhibit 39:

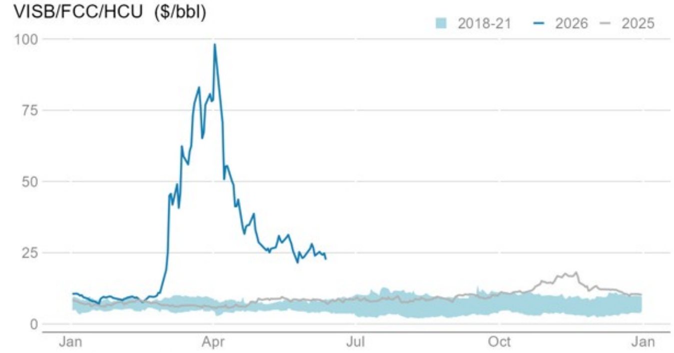
Cracking netback margin - Southeast Asia
Topping ($/bbl)



Source: S&P Global Platts, Morgan Stanley Research

Exhibit 40:

Cracking netback margin - Southeast Asia
VISB/FCC/HCU ($/bbl)



Source: S&P Global Platts, Morgan Stanley Research

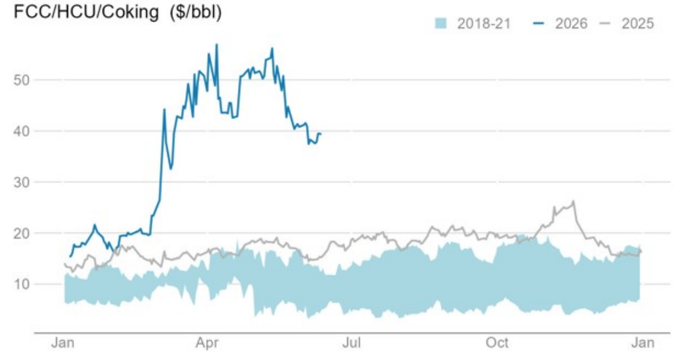
Exhibit 41:

Cracking netback margin - Western Mediterranean
VISB/FCC/HCU ($/bbl)

Source: S&P Global Platts, Morgan Stanley Research

Exhibit 42:

Cracking netback margin - USGC
FCC/HCU/Coking ($/bbl)



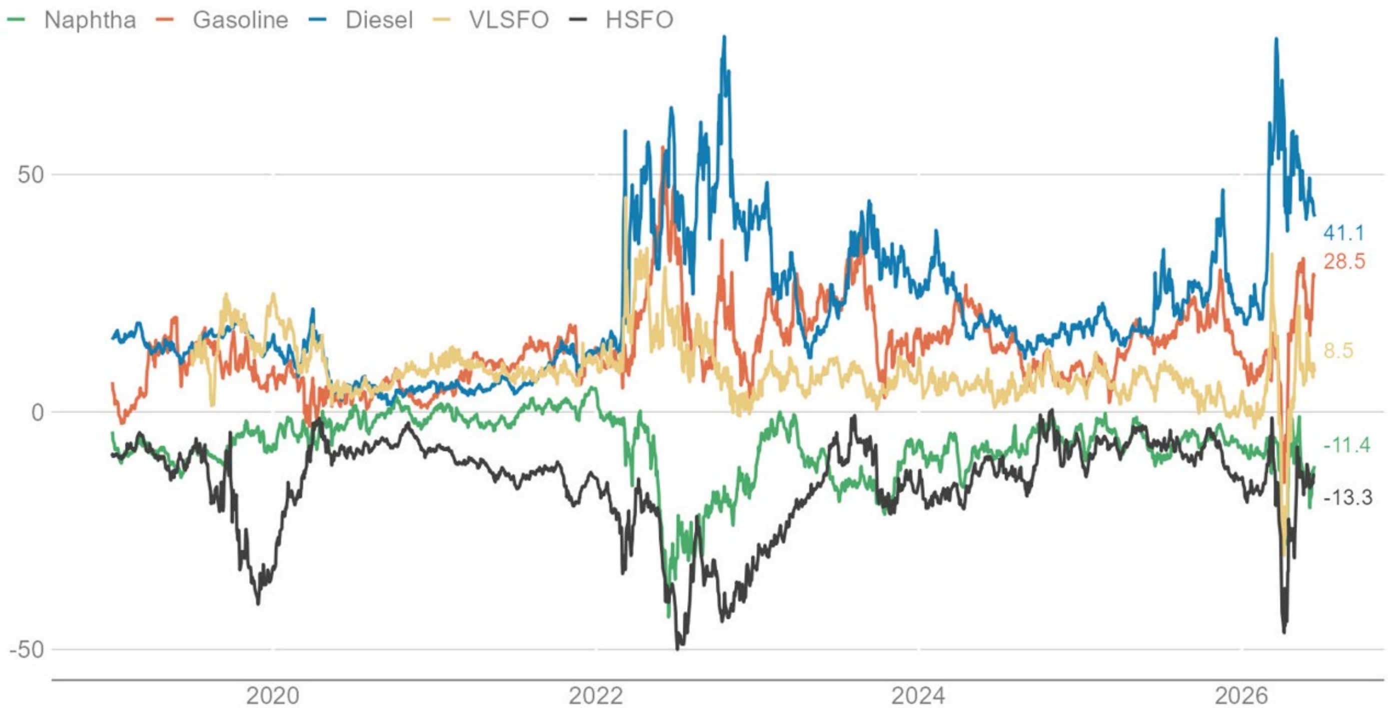
Source: S&P Global Platts, Morgan Stanley Research

Product Crack Spreads

Exhibit 43:

Refined product crack spreads

vs Dated Brent; basis: Antwerp-Rotterdam-Amsterdam ($/bbl)

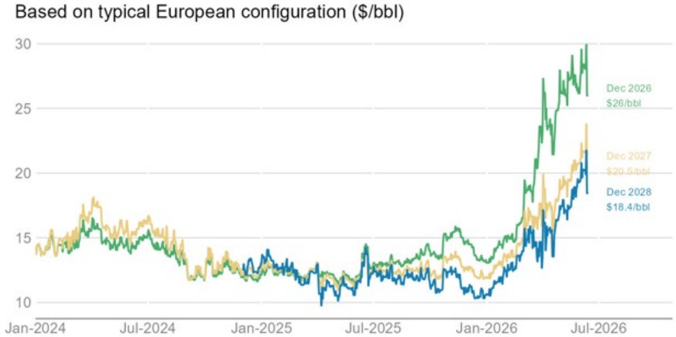


Source: Platts, Morgan Stanley

Exhibit 44:

Forward refining margins

Based on typical European configuration ($/bbl)

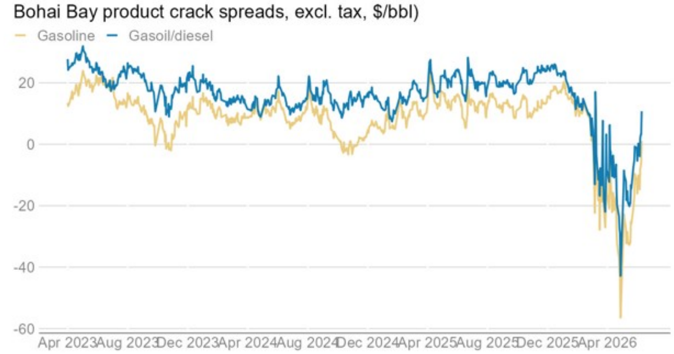


Note: Analysis assumes yields of 30% gasoline, 60% middle distillate, 5% naphtha and 5% HSFO
Source: Bloomberg, Morgan Stanley Research

Exhibit 45:

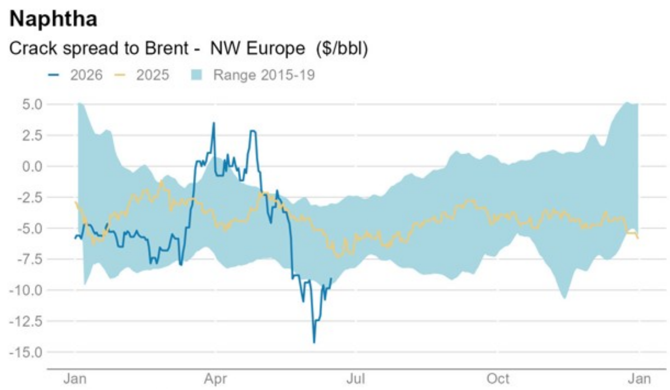
China

Bohai Bay product crack spreads, excl. tax, $/bbl)



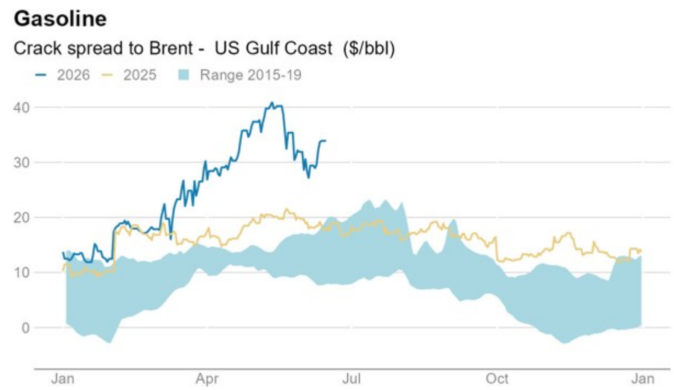
Source: Argus, Morgan Stanley Research

Exhibit 46:



Source: S&P Global Platts, Morgan Stanley Research

Exhibit 47:



Source: S&P Global Platts, Morgan Stanley Research

Exhibit 48:

Source: S&P Global Platts, Morgan Stanley Research

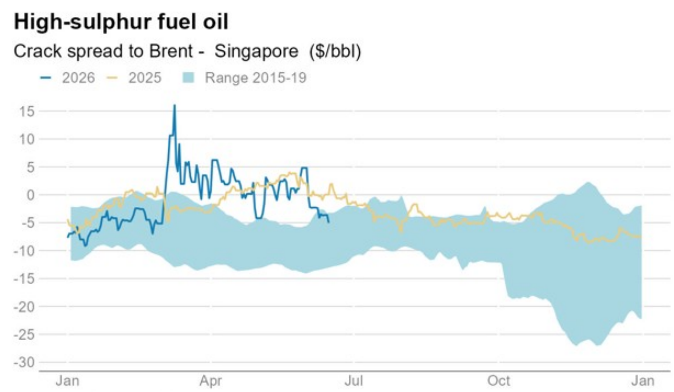
Exhibit 49:

Source: S&P Global Platts, Morgan Stanley Research

Exhibit 50:

Source: S&P Global Platts, Morgan Stanley Research

Exhibit 51:



Source: S&P Global Platts, Morgan Stanley Research

Refinery Outages

Exhibit 52:

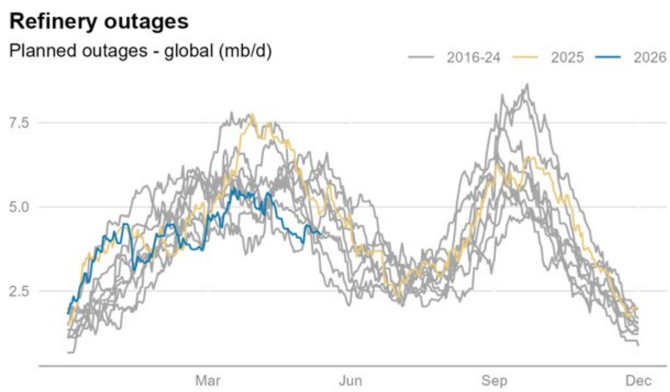
Refinery outages

(mb/d)

	Actual													Known outlook	
region	May 2025	Jun 2025	Jul 2025	Aug 2025	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mar 2026	Apr 2026	May 2026	Jun 2026	Jul 2026
Africa	0.70	0.60	0.47	0.67	0.60	0.38	0.37	0.36	0.61	0.56	0.27	0.13	0.09	0.11	0.13
Brazil	0.11	0.14	0.16	0.09	0.09	0.12	0.07	0.07	0.05	0.04	0.05	0.04	0.04	0.01	0.02
China	2.10	1.51	1.20	0.91	0.86	0.96	1.22	1.19	1.12	0.53	0.82	1.39	1.82	2.03	1.24
Europe	1.51	1.12	0.38	0.17	0.57	1.03	1.16	0.36	0.31	0.80	1.20	0.69	0.61	0.34	0.10
India	0.15	0.20	0.08	0.26	0.35	0.23	0.27	0.20	0.09	0.10	0.18	0.52	0.56	0.53	0.35
Japan	0.60	0.79	0.52	0.35	0.42	0.46	0.42	0.10	0.13	0.15	0.25	0.44	0.40	0.48	0.33
Korea	0.08	0.12	0.00	0.00	0.00	0.17	0.17	0.17	0.00	0.24	0.38	0.64	0.53	0.47	0.03
Middle East	0.51	0.40	0.11	0.16	0.21	0.77	1.19	0.54	0.45	0.41	1.55	1.88	1.73	1.32	0.25
Other	2.84	2.12	1.73	1.59	1.91	2.08	1.37	1.11	1.26	1.66	2.48	3.26	3.08	2.06	0.78
Russia	0.60	0.72	0.92	1.48	2.05	2.21	1.98	1.50	0.72	0.78	0.98	2.01	2.29	1.63	0.39
Taiwan	0.16	0.00	0.00	0.00	0.24	0.24	0.24	0.31	0.15	0.00	0.18	0.32	0.16	0.14	0.15
US	0.75	0.31	0.29	0.32	0.83	1.55	0.63	0.19	0.88	1.37	0.95	1.10	0.43	0.15	0.14
World	**10.11**	**7.97**	**5.86**	**5.94**	**7.95**	**10.11**	**9.09**	**5.84**	**5.69**	**6.43**	**9.27**	**12.42**	**11.76**	**8.97**	**3.63**

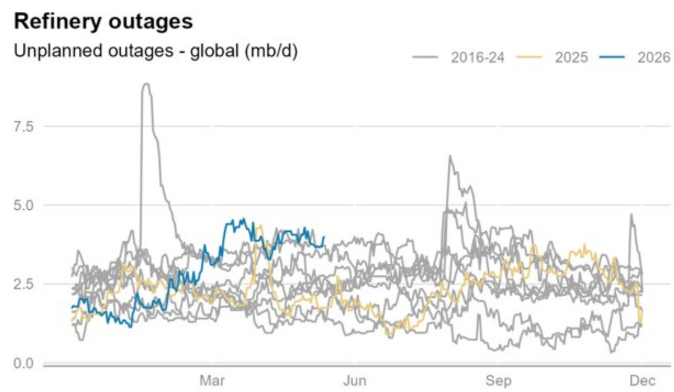
Source: IIR, Morgan Stanley Research

Exhibit 53:



Source: IIR, Morgan Stanley Research

Exhibit 54:



Source: IIR, Morgan Stanley Research

Inventories

Exhibit 55:

Observable crude oil inventories

(million bbl)

		Level	Average daily change					
		Current	-4 wks	-13 wks	-26 wks	May	Apr	Mar
Commercial								
	United States	424.9	−0.85	−0.24	0.03	−0.85	−0.08	0.73
	Japan	58.1	0.35	−0.06	−0.06	0.39	−0.15	−0.39
	ARA region	55.3	0.26	0.06	0.01	0.00	−0.19	0.33
	Other OECD	549.9	0.34	−0.36	−0.14	−0.12	−0.42	−0.25
	China	1,111.2	−0.25	0.20	0.54	−0.86	0.58	1.26
	Other non-OECD	1,136.9	0.01	−0.20	0.05	0.03	−0.67	0.32
	Floating storage	81.7	−0.21	−0.06	−0.27	−1.72	−0.48	1.18
	Oil-in-transit	1,132.0	−0.21	−0.02	−0.37	−0.20	1.32	−3.61
Sub-total	—	**4,550.1**	**−0.6**	**−0.7**	**−0.2**	**−3.3**	**−0.1**	**−0.4**
SPR								
	United States	349.2	−1.25	−0.73	−0.34	−1.32	−0.57	−0.01
	Japan	262.2	−0.13	0.01	0.01	0.08	−0.04	−0.02
	ARA region	1.0	0.00	0.00	0.00	0.00	0.00	0.00
	Other OECD	257.0	0.00	0.01	−0.01	0.00	0.00	0.03
	China	188.8	−0.06	0.00	0.01	0.05	0.07	−0.07
Sub-total	—	**1,058.2**	**−1.4**	**−0.7**	**−0.3**	**−1.2**	**−0.5**	**−0.1**
Total	—	**5,608.3**	**−2.0**	**−1.4**	**−0.5**	**−4.5**	**−0.6**	**−0.5**

Source: IEA, EIA/DOE, PJK, IE, Genscape, PAJ, Platts, Vortexa, Morgan Stanley Research

Exhibit 56:

Observable oil products inventories

(million bbl)

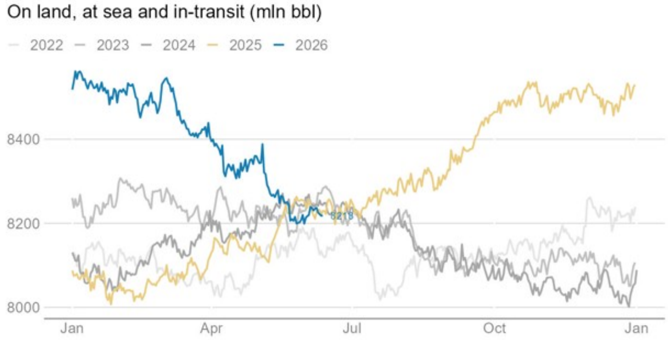
		Level	Average daily change					
		Current	-4 wks	-13 wks	-26 wks	May	Apr	Mar
On land								
	Light ends	428.9	−0.08	−0.56	−0.07	−0.29	−0.86	−0.65
	Middle distillates	688.1	−0.07	−0.24	−0.12	−0.07	−0.54	−0.25
	Fuel oil & resid	117.2	−0.23	−0.25	−0.19	−0.07	−0.29	−0.25
	Other/not specified	566.1	−0.04	0.09	−0.27	0.26	0.27	−0.23
Sub-total	—	**1,800.34**	**−0.42**	**−0.96**	**−0.65**	**−0.17**	**−1.42**	**−1.38**
At sea								
	Light ends	180.3	−0.36	−0.16	−0.12	−0.68	0.32	−0.58
	Middle distillates	331.1	0.17	−0.26	−0.05	−0.04	−0.22	−0.89
	Fuel oil & resid	126.9	0.14	−0.04	−0.06	−0.15	−0.01	−0.40
	Other/not specified	171.2	0.30	0.30	0.09	0.37	0.40	−0.26
Sub-total	—	**809.44**	**0.25**	**−0.16**	**−0.14**	**−0.50**	**0.49**	**−2.13**
Total	—	**2,609.79**	**−0.17**	**−1.12**	**−0.79**	**−0.67**	**−0.93**	**−3.51**

Source: IEA, EIA/DOE, PJK, IE, Genscape, PAJ, Platts, Vortexa, Morgan Stanley Research

Exhibit 57:

Observable crude oil and oil products inventories

On land, at sea and in-transit (mln bbl)

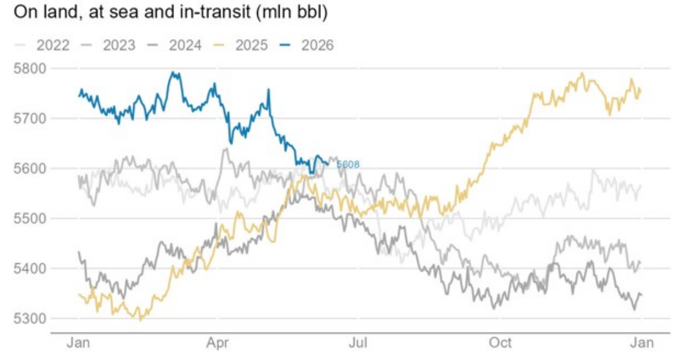


Note: Inventories include SPR
Source: IEA, EIA/DOE, PJK, IE, Genscape, PAJ, Platts, Vortexa, Morgan Stanley Research

Exhibit 58:

Observable crude oil inventories

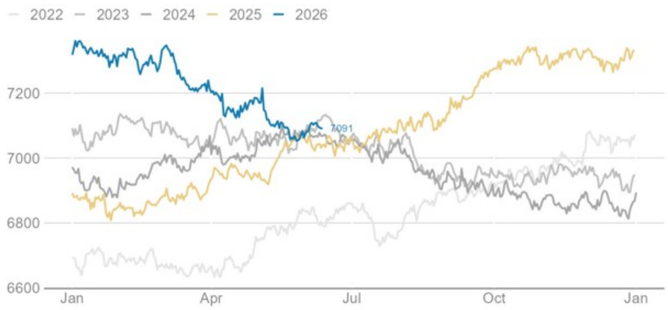
On land, at sea and in-transit (mln bbl)



Source: IEA, EIA/DOE, PJK, IE, Genscape, PAJ, Platts, Vortexa, Morgan Stanley Research

Exhibit 59:

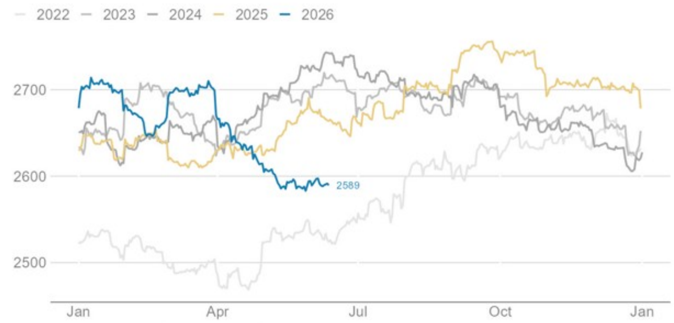
Observable crude oil and oil products inventories
In commercial storage only (mln bbl)



Note: inventories on land, at sea and in-transit, not including SPR in OECD countries
Source: IEA, EIA/DOE, PJK, IE, Genscape, PAJ, Platts, Vortexa, Morgan Stanley Research

Exhibit 60:

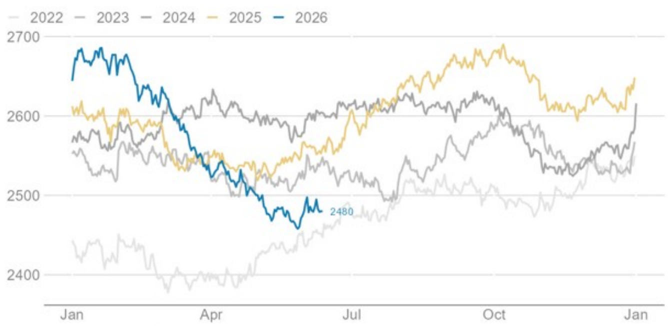
OECD commercial oil inventories
Total oil, on land only (mln bbl)



Source: IEA, EIA/DOE, PJK, PAJ, Platts, Vortexa, Morgan Stanley Research

Exhibit 61:

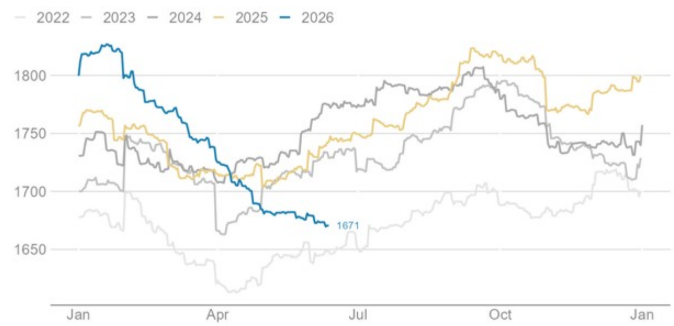
Observable refined oil products inventories
On land, at sea and in-transit (mln bbl)



Source: IEA, EIA/DOE, PJK, IE, PAJ, Platts, Morgan Stanley Research

Exhibit 62:

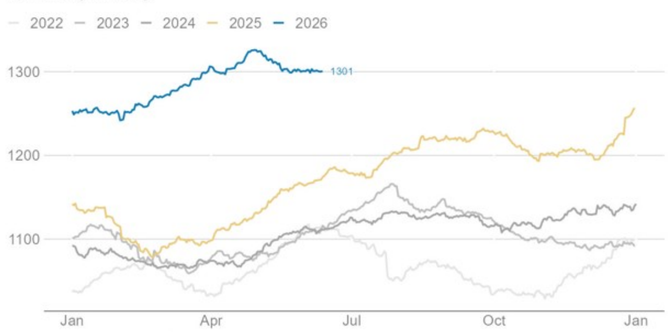
Observable refined oil products inventories, on-land only
(mln bbl)



Source: IEA, EIA/DOE, PJK, IE, PAJ, Platts, Morgan Stanley Research

Exhibit 63:

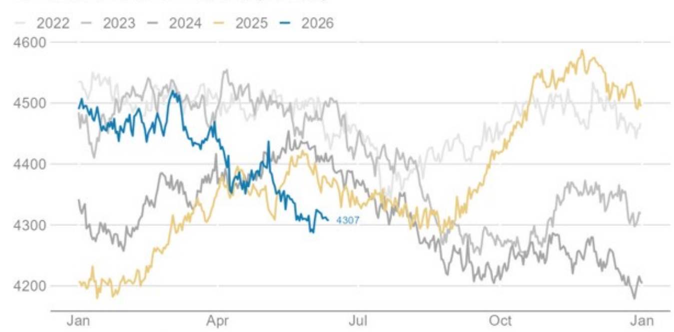
Observable crude oil inventories in China
On land (mln bbl)



Source: Vortexa, Morgan Stanley Research

Exhibit 64:

Observable crude oil inventories outside China
On land, at sea and in-transit (mln bbl)



Source: IEA, EIA/DOE, PJK, IE, PAJ, Platts, Morgan Stanley Research

Seaborne Exports

Exhibit 65:

OPEC 9+3

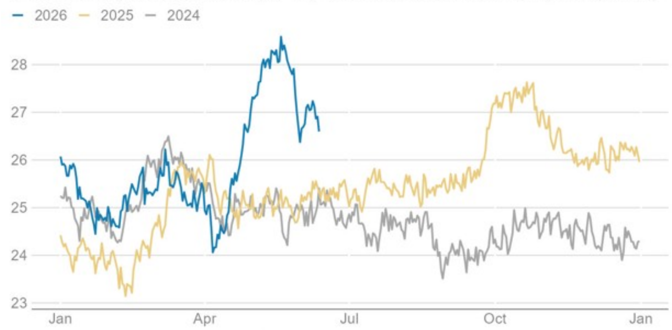
Seaborne exports of crude oil and oil products (mb/d)

Source: Vortexa, Morgan Stanley Research

Exhibit 66:

Non-OPEC

Seaborne exports of crude oil (30-day mov. avg; excl. intra-country flows; mb/d)



Source: Vortexa, Morgan Stanley Research

Exhibit 67:

Saudi Arabia

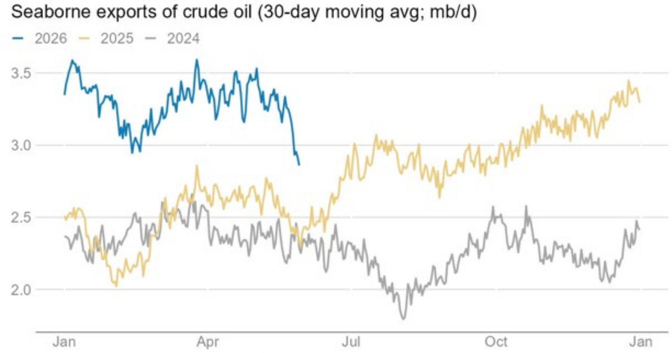
Seaborne crude oil exports (mb/d)

Source: Vortexa, Morgan Stanley Research

Exhibit 68:

Brazil and Guyana

Seaborne exports of crude oil (30-day moving avg; mb/d)



Source: Vortexa, Morgan Stanley Research

Exhibit 69:

Russia

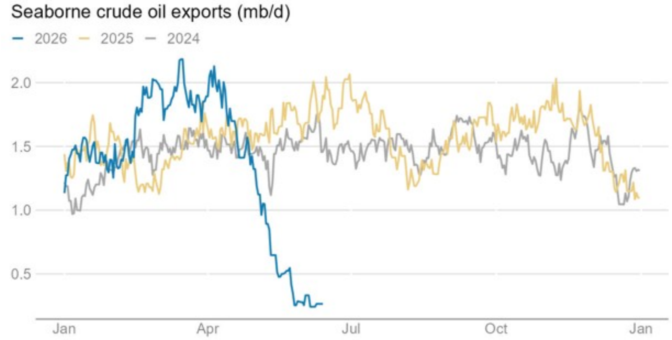
Seaborne crude oil exports (mb/d)

Source: Vortexa, Morgan Stanley Research

Exhibit 70:

Iran

Seaborne crude oil exports (mb/d)



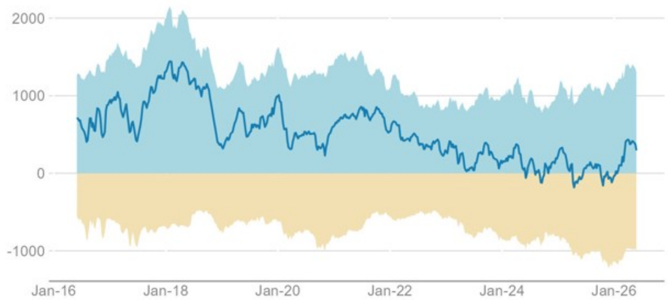
Note: chart shows data to 13 Jun
Source: Vortexa, Morgan Stanley Research

Positioning

Exhibit 71:

Total oil

COT report - Managed money and Other reportables (number of contracts)

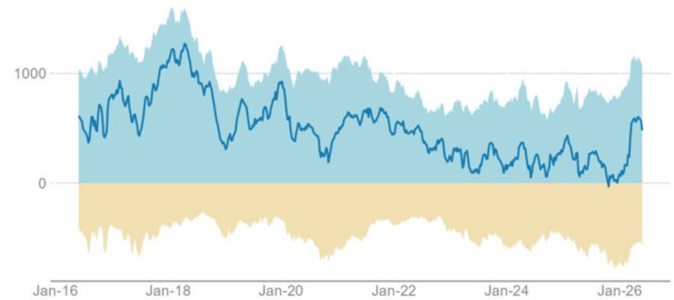


Note: light blue = long positions, yellow = short postion, dark blue = net position
Source: CFTC, Morgan Stanley Research

Exhibit 72:

Brent and WTI

COT report - Managed money and Other reportables (number of contracts)

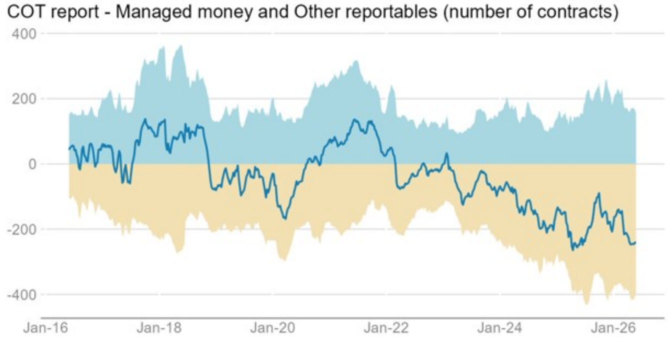


Note: light blue = long positions, yellow = short postion, dark blue = net position
Source: CFTC, Morgan Stanley Research

Exhibit 73:

Gasoil and Heating Oil

COT report - Managed money and Other reportables (number of contracts)

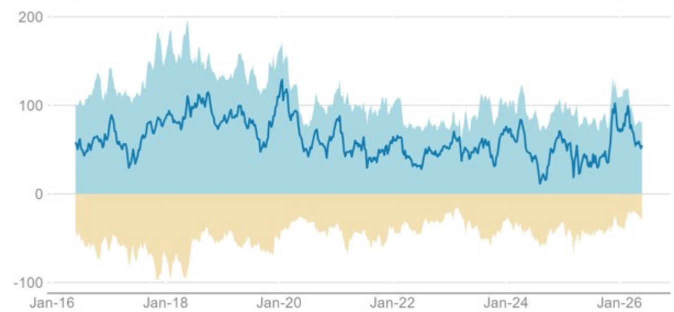


Note: light blue = long positions, yellow = short postion, dark blue = net position
Source: CFTC, Morgan Stanley Research

Exhibit 74:

Gasoline - NY Harbour RBOB

COT report - Managed money and Other reportables (number of contracts)



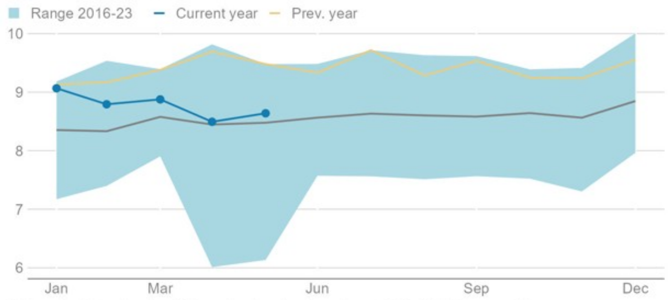
Note: light blue = long positions, yellow = short postion, dark blue = net position
Source: CFTC, Morgan Stanley Research

Other Selected Charts

Exhibit 75:

Motor fuel imports

Top 100 ports (mb/d)

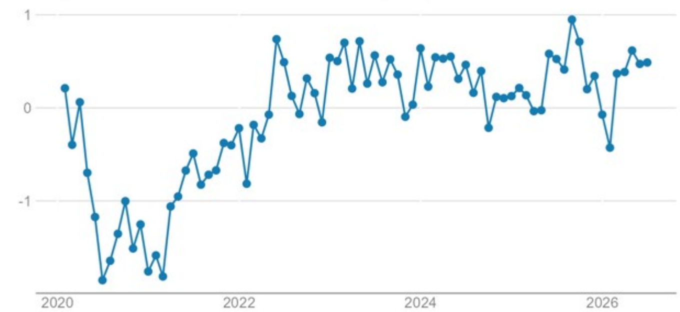


Note: analysis based on the 100 largest ports where imports are >90% of total flow, excluding large trading ports like Singapore, Rotterdam, etc
Source: Vortexa, Morgan Stanley Research

Exhibit 76:

Seaborne crude oil arrivals - Europe

Change vs seasonal average from 2016-23 (mb/d)

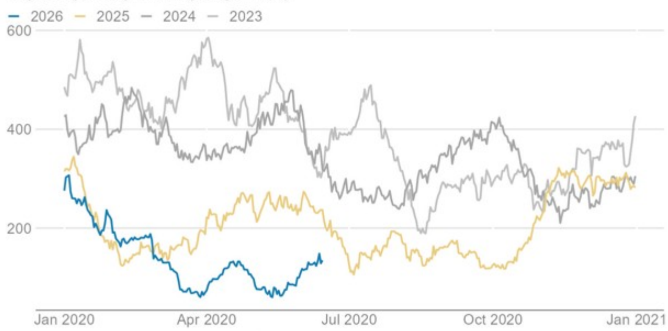


Note: analysis based on EU + Norway + UK
Source: Vortexa, Morgan Stanley Research

Exhibit 77:

Clean products net imports

Nigeria (30-day moving avg; mb/d)

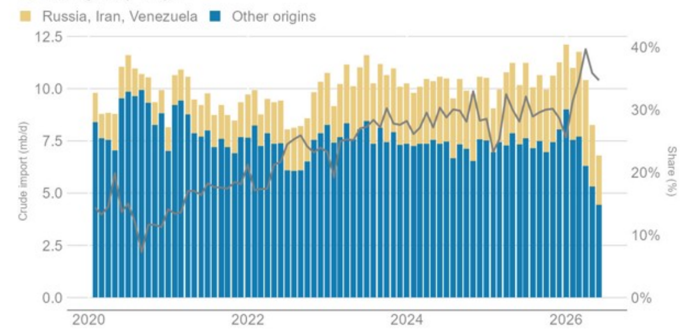


Source: Vortexa, Morgan Stanley Research

Exhibit 78:

Crude oil imports

China, split by origin

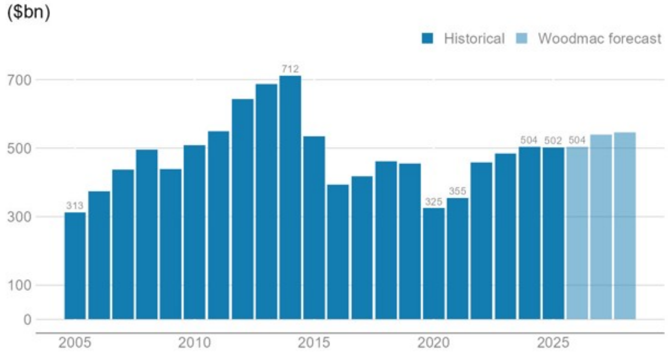


Source: Vortexa, Morgan Stanley Research

Exhibit 79:

Global oil & gas capex (by Woodmac)

($bn)

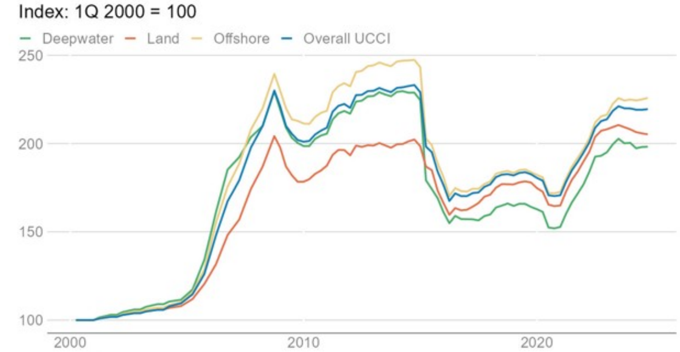


Source: Wood Mackenzie, Morgan Stanley Research

Exhibit 80:

Upstream capital cost

Index: 1Q 2000 = 100



Source: S&P Global, Morgan Stanley Research

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(as of May 31, 2026)

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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
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Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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